RAJALAKSHMI ENGINEERING COLLEGE

Muralikarthik V 250701458-2025-CSE

M2

Started on	Friday, 21 August 2026, 4:30 PM
State	Finished
Completed on	Friday, 25 September 2026, 3:53 PM
Time taken	34 days 23 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```c
#include <stdio.h>
void quicksort(int arr[], int low, int high, int n) {
    if (low < high) {
        int pivot = arr[low];
        int i = low + 1;
        int j = high;
        while (i <= j) {
            while (i <= high && arr[i] <= pivot) {
                i++;
            }
            while (j >= low && arr[j] > pivot) {
                j--;
            }
            if (i < j) {
                int temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
            }
        }
        int temp = arr[low];
        arr[low] = arr[j];
        arr[j] = temp;
        quicksort(arr, low, j - 1, n);
        quicksort(arr, j + 1, high, n);
    }
}
int main() {
    int n;
    scanf("%d", &n);
    int arr[n];
    for (int i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }
    quicksort(arr, 0, n - 1, n);
    for (int i = 0; i < n; i++)
    {
        printf("%d ", arr[i]);
    }
    printf("\n");
    return 0;
}
```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔

	Input	Expected	Got	
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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